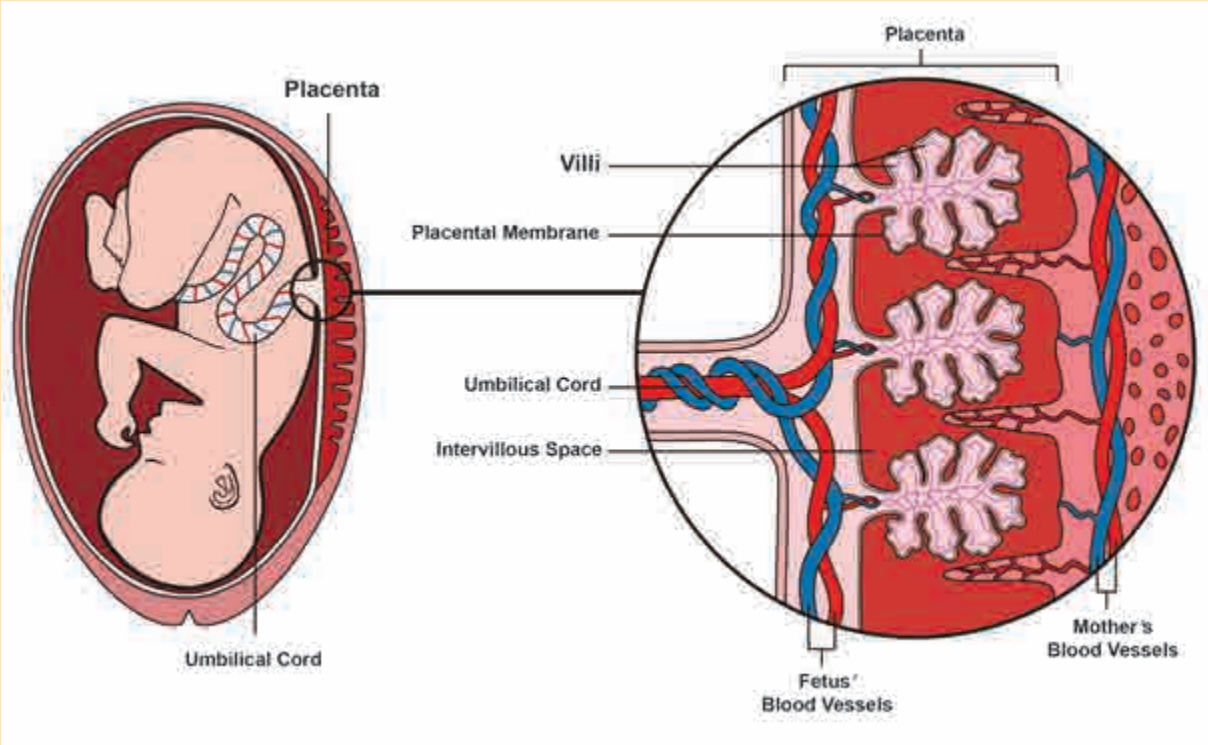
www.tntextbooks.in

DO YOU KNOW?

How does oxygen exchanges between maternal and fetal blood in a pregnant women?



In a pregnant women, the oxygen supply for fetus is provided by the maternal blood in the placenta where the blood vessels of both mother and fetus are in close proximity. Both fetal and maternal hemoglobin binds to oxygen reversibly as follows.

$$Hb(mother) + O_2 \rightleftharpoons HbO_2\ (mother)$$

$$Hb(fetus) + O_2 \rightleftharpoons HbO_2\ (Fetus)$$

In the above two equilibriums, the equilibrium constant value for the oxygenation of fetal hemoglobin is higher, which is due to its higher affinity for oxygen compared to the adult hemoglobin. Hence in placenta, the oxygen from the mother's blood is effectively transferred to the fetal hemoglobin

SUMMARY

- In our daily life, we observe several chemical and physical changes.
- In some chemical reactions, at a particular stage, the rate of the reverse reaction is equal to that of the forward reaction indicating a state of equilibrium.